

Assignment: Build an Inventory & Logistics Management Web App for Campus Express

Objective:

Your goal is to build a functional prototype of an **Inventory & Logistics Management System** for Campus Express that includes (but is not limited to) the following features/components:

Requirements:

Front-end

- Dashboard for operations managers showing key metrics:
 - Total shipments in transit, delivered, delayed
 - Warehouse inventory status (packaging material, empty boxes, labels, etc.)
 - Alerts/notifications: low stock, delayed packages, etc.
- Shipment management module:
 - Create new shipment orders (sender, receiver, origin, destination, status updates)
 - View list of shipments, searchable/filterable by date, status, etc.
 - Update shipment status (e.g. "Picked up", "In transit", "Out for delivery", "Delivered")
- Inventory module:
 - Add or update items (packaging supplies etc.)
 - Track stock levels, show alerts when below threshold
- Warehouse module: mapping of warehouses (locations), their current capacity and inventory

Back-end

- RESTful API endpoints for front-end to consume (CRUD operations for shipments, inventory items, warehouses, users)
- Database design: entities like Shipment, InventoryItem, Warehouse, User, etc., with relations
- Authentication & authorization: different roles (e.g. Admin, Manager, Warehouse Staff) with different permissions
- Real-time updates (optional/bonus): e.g. WebSocket or polling to update dashboard when shipment status changes

Additional Features (Optional / Bonus)

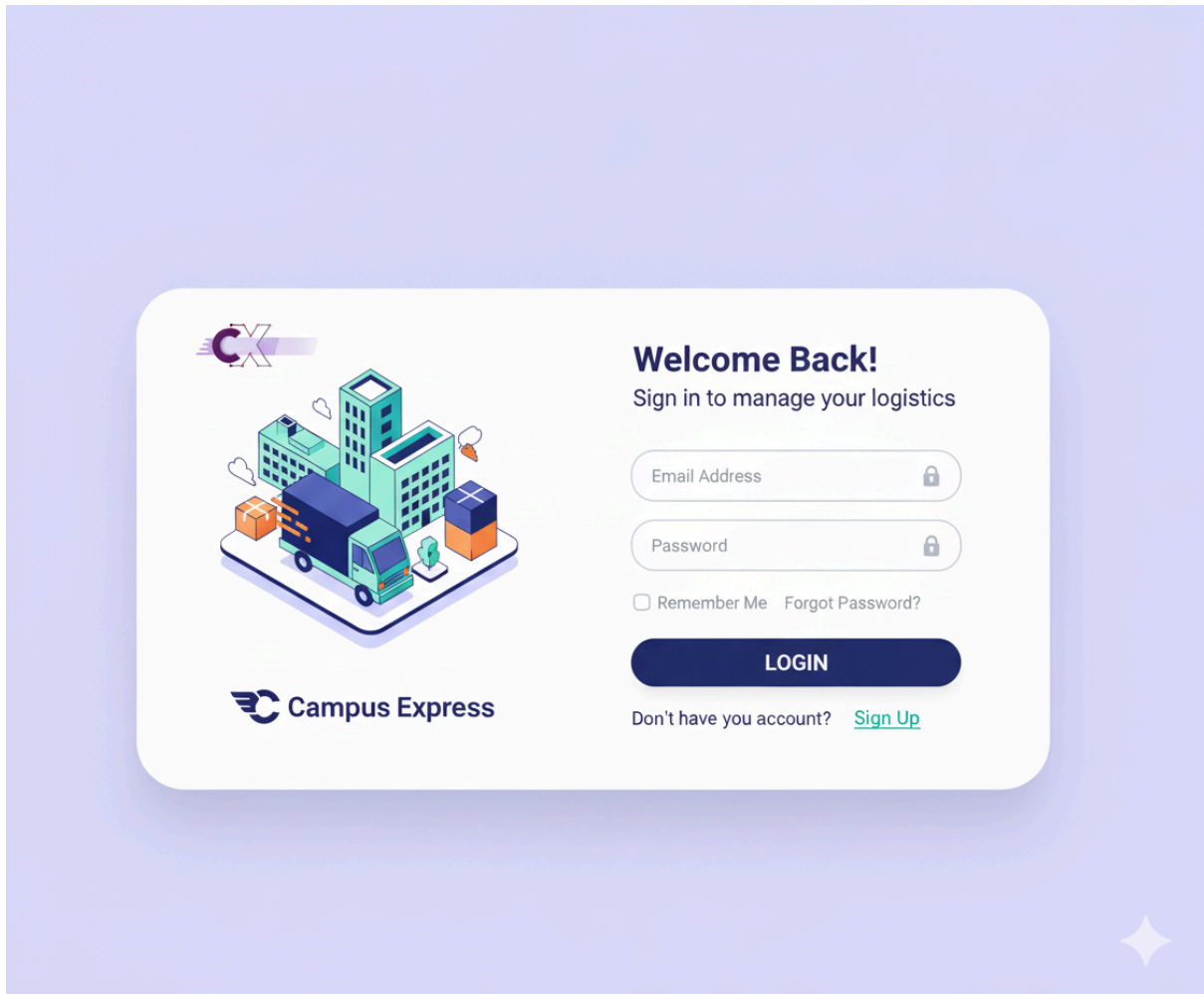
- Barcode / QR code scanning for packages or inventory items
- Geolocation/tracking integration for shipments
- Notifications via email or SMS on shipment status changes or low stock
- Mobile-friendly / responsive UI
- Data import/export (CSV, Excel)

Assignment Deliverables:

- Source code (front-end + back-end)
- A working deployment (local or hosted in render,vercel,heroku,etc.[Bonus])
- Screenshots or screen recordings of the app in action

Template examples:

A) Login/Signup Page:



The image shows a login/signup page template for 'Campus Express'. The page has a light purple background. On the left, there is an illustration of a city with buildings, a delivery truck, and boxes. Below this illustration is the 'Campus Express' logo, which consists of a stylized 'C' with wings and the text 'Campus Express'. To the right of the illustration, the text 'Welcome Back!' is displayed in a bold, dark blue font, followed by the subtitle 'Sign in to manage your logistics'. Below this, there are two input fields: 'Email Address' and 'Password', both with lock icons on the right. Under the 'Password' field, there is a checkbox labeled 'Remember Me' and a link 'Forgot Password?'. A dark blue button with the text 'LOGIN' in white is positioned below these fields. At the bottom left, there is a link 'Don't have you account? [Sign Up](#)'. A small white star icon is located in the bottom right corner of the page.



Welcome Back!
Sign in to manage your logistics

Email Address 

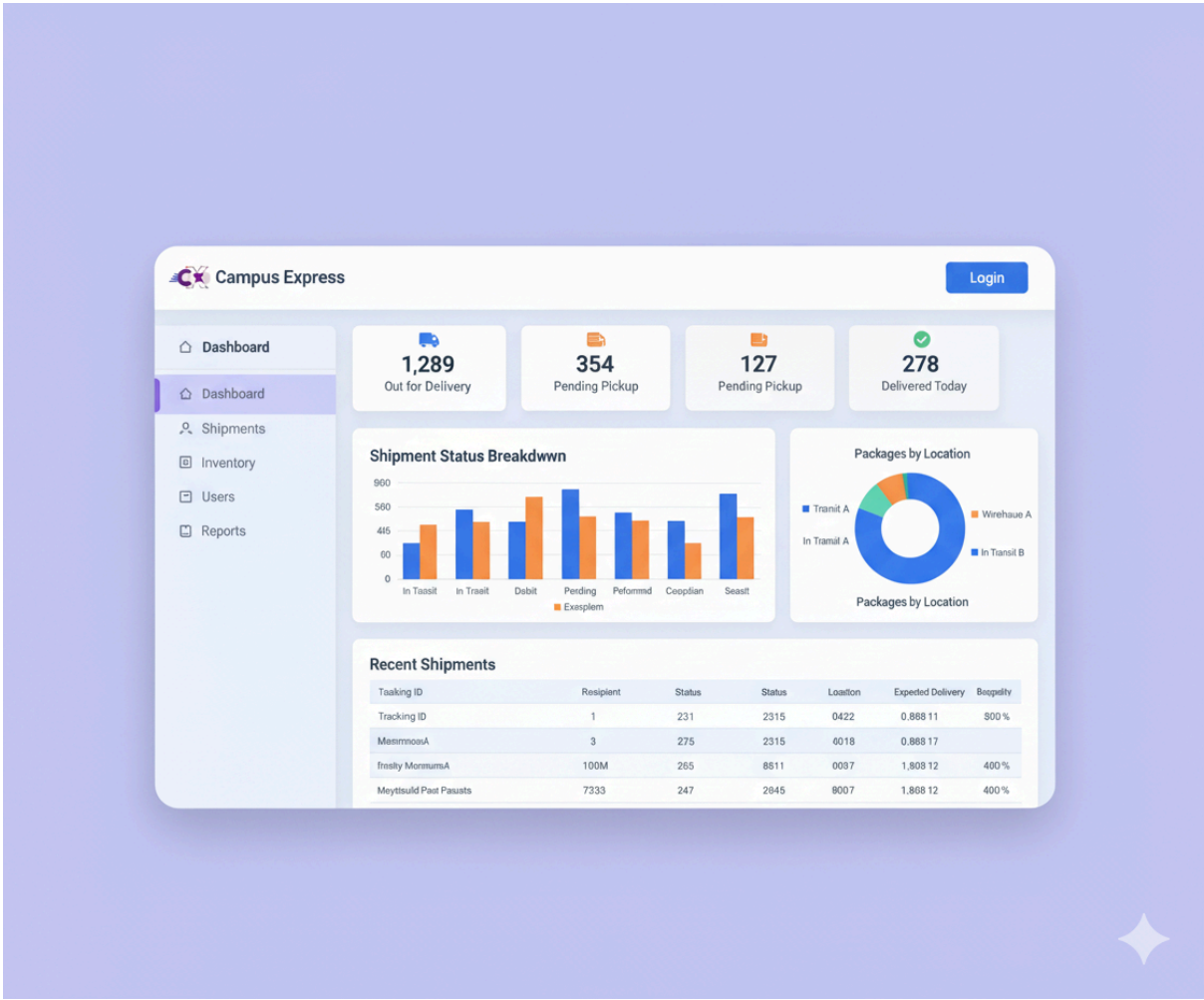
Password 

☐ Remember Me [Forgot Password?](#)

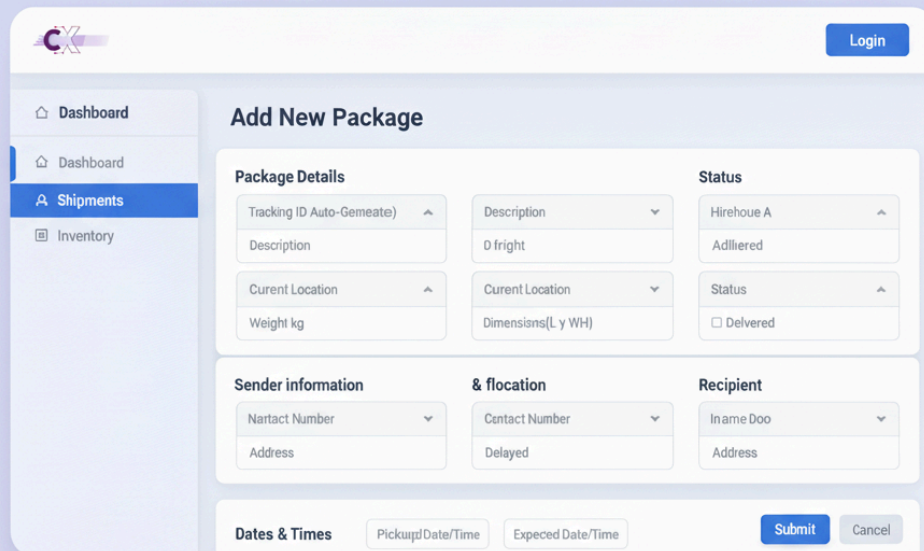
LOGIN

Don't have you account? [Sign Up](#)

B) Dashboard Overview:



C) Add/edit package form(Shipments) :



The screenshot shows a web application interface for adding a new package. The main heading is "Add New Package". The form is organized into several sections:

- Package Details:** Includes fields for "Tracking ID Auto-Generate" (with an up arrow), "Description" (with a dropdown), "Current Location" (with an up arrow), "Weight kg", "Description" (with a dropdown), "Current Location" (with a dropdown), and "Dimensions(L x W x H)".
- Status:** Includes fields for "Warehouse A" (with an up arrow), "Assigned" (with a dropdown), "Status" (with an up arrow), and a checkbox for "Delivered".
- Sender information:** Includes fields for "Contact Number" (with a dropdown) and "Address".
- & location:** Includes fields for "Contact Number" (with a dropdown) and "Delayed".
- Recipient:** Includes fields for "Name Doo" (with a dropdown) and "Address".
- Dates & Times:** Includes fields for "Pickup Date/Time" and "Expected Date/Time".

At the bottom right, there are "Submit" and "Cancel" buttons. The left sidebar shows a navigation menu with "Dashboard", "Shipments", and "Inventory". The top right corner has a "Login" button.

D) User Management :

